

6.2.1 Probing deep into matter

Learning outcomes	Additional guidance
(a) <i>Describe and explain:</i>	
(i) use of particle accelerators to generate high-energy beams of particles for scattering	HSW7, 11
(ii) evidence from scattering for a small massive nucleus within the atom	Development of the nuclear model HSW7, HSW11
(iii) evidence of discrete energy levels in atoms	For example from collisions with electrons or from line spectra, HSW12
(iv) a simple model of the atom as the quantum behaviour of electrons in a confined space	HSW7
(v) simple picture of the quark structure of protons and neutrons	
(vi) application of conservation of mass/energy, charge and lepton number in balanced nuclear equations	
(vii) relativistic calculations for particles travelling at very high speed, for example in particle accelerators or cosmic rays.	
(b) <i>Make appropriate use of:</i>	
(i) the terms: energy level, scattering, nucleus, proton, neutron, nucleon, electron, positron, quark, gluon, neutrino, hadron, lepton, antiparticle, lepton number	HSW8
<i>by sketching and interpreting:</i>	
(ii) paths of scattered particles	
(iii) electron standing waves in simple models of an atom.	

(c) *Make calculations and estimates involving:*

- (i)** motion of a charged particle in magnetic field using $F = qvB$ *M2.3*
- (ii)** kinetic and potential energy of a scattered charged particle
- (iii)** $E_{\text{rest}} = mc^2$ and relativistic factor *M2.2*
HSW11, 12
- (iv)** $E_{\text{total}} = \gamma E_{\text{rest}}$.